

5. Equilibrium of a Rigid Body

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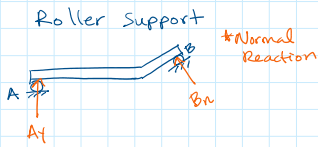
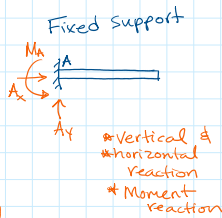
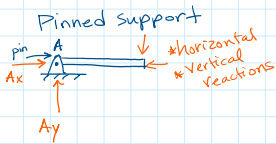
- * When a body is in a state of Equilibrium its acceleration is zero.
 - * Rigid Body - a body that does not deform (shape is not altered) due to forces being exerted on it.
 - * Sum of Forces equals zero $\sum F = 0$
 - * Sum of Moments equals zero $\sum M = 0$
- New $\rightarrow$

In 2D $\rightarrow \sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_o = 0$
 $\rightarrow \sum F_{Ax} = 0 \quad \sum F_{Ay} = 0 \quad \sum M_o = 0$

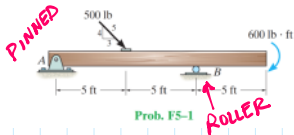


$$F_s = k(l - l_o) \text{ - Spring Force}$$

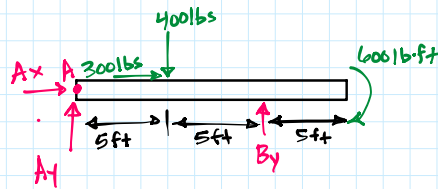
Types of Supports



F5-1. Determine the horizontal and vertical components of reaction at the supports. Neglect the thickness of the beam.



FBD on the Beam



Draw reaction forces
 Draw external forces
 & moments

$$\rightarrow \sum F_x = 0 \rightarrow Ax + 300 = 0$$

$$Ax = -300$$

$$Ax = 300 \text{ lbs} \leftarrow$$

$$\uparrow \sum F_y = 0$$

$$Ay + By - 400 = 0$$

$$Ay = 400 - By$$

$$Ay = 400 - 260$$

$$Ay = 140 \text{ lbs} \uparrow$$

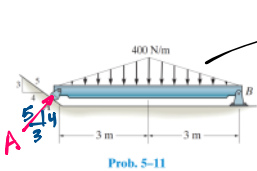
$$\uparrow \sum M_A = 0$$

$$-(400 \text{ lbs})(5 \text{ ft}) + By(10 \text{ ft}) - 600 \text{ lb} \cdot \text{ft} = 0$$

$$10 By = 2600$$

$$By = 260 \text{ lbs} \uparrow$$

5-11. Determine the reactions at the supports.

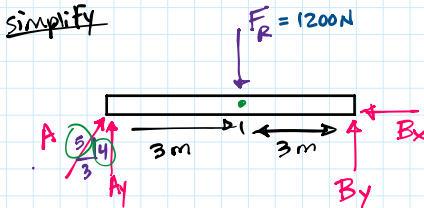


$$F_R = \int_L w dx = \frac{1}{2} b h$$

$$F_R = \frac{1}{2} (6 \text{ m}) (400 \frac{\text{N}}{\text{m}}) = 1200 \text{ N}$$

$$Bx = \frac{3}{5} (750) = 450 \text{ N} \leftarrow$$

$$\sum F_x = 0$$



$$\uparrow \sum M_B = 0$$

$$(3 \text{ m})(1200 \text{ N}) - (6 \text{ m})(\frac{4}{5} A) = 0$$

$$A = 750 \text{ N} \quad \frac{5}{3} \uparrow \text{ or } \angle 53^\circ$$

$$B_x = \frac{3}{5}(750) = 450 \text{ N} \leftarrow$$

$$A = 750 \text{ N} \frac{5}{14} \text{ or } \angle 53^\circ$$

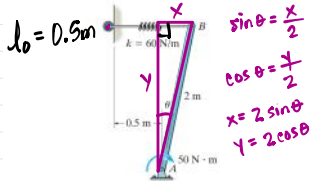
$$+\uparrow \Sigma F_y = 0 \quad A_y + B_y - 1200 = 0$$

$$B_y = 1200 - \frac{4}{5}(750) = 1200 - 600 = 600 \text{ N}$$

$$B_y = 600 \text{ N} \uparrow$$

$$B_y = 1200 - A_y$$

*5-44. The 10-kg uniform rod is pinned at end A. If it is also subjected to a couple moment of 50 N·m, determine the smallest angle θ for equilibrium. The spring is unstretched when $\theta = 0$, and has a stiffness of $k = 60 \text{ N/m}$.



$$m = 10 \text{ kg}$$

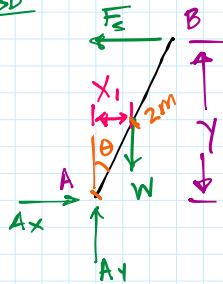
$$M = 50 \text{ N}\cdot\text{m} \quad \text{CW}$$

$$\theta = ?$$

$$K = 60 \text{ N/m} \quad W = mg = 10(9.81) = 98.1 \text{ N}$$

$$\Sigma \vec{F} = 0 \quad \Sigma M_o = 0$$

FBD



$$\sin \theta = \frac{x_1}{1}$$

$$x_1 = 1 \sin \theta$$

$$F_s = k(l - l_0) = k \cdot x$$

$$\Sigma F_x = 0 \rightarrow A_x - F_s = 0$$

$$A_x = F_s$$

$$\Sigma F_y = 0 \rightarrow A_y - W = 0$$

$$A_y = W$$

$$+\circlearrowleft \Sigma M_A = -50 - x_1 W + y F_s = 0$$

$$-50 - \sin \theta (98.1) + 2 \cos \theta (K \cdot 2 \sin \theta) = 0$$

$$-50 - 98.1 \sin \theta + 240 \cos \theta \sin \theta = 0$$

$$-5 - 9.81 \sin \theta + 24 \cos \theta \sin \theta = 0$$

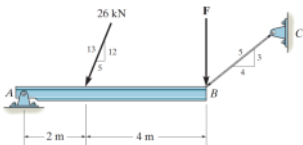
$$5 = 24 \cos \theta \sin \theta - 9.81 \sin \theta$$

Solve graphically

$$\theta = 0.429 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right)$$

$$\theta = 24.6^\circ$$

*5-48. If rope BC will fail when the tension becomes 50 kN, determine the greatest vertical load F that can be applied to the beam at B. What is the magnitude of the reaction at A for this loading? Neglect the thickness of the beam.



Probs. 5-47/48

$$+\circlearrowleft \Sigma M_A = -2(24) - 6(F) + 6(30) = 0$$

$$-48 + 180 = 6F$$

$$\frac{132}{6} = \frac{6F}{6}$$

$$F = 22 \text{ kN} \downarrow$$

$$\Sigma F_x = A_x - 10 + 40 = 0$$

$$A_x = -30 \text{ kN}$$

$$A_x = 30 \text{ kN} \leftarrow$$

$$\Sigma F_y = A_y - 24 - F + 30 = 0$$

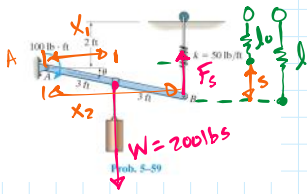
$$A_y = 24 + 22 - 30$$

$$A_y = 16 \text{ kN} \uparrow$$

5-59. The rod supports a weight of 200 lb and is pinned at its end A. If it is also subjected to a couple moment of 100 lb·ft, determine the angle θ for equilibrium. The spring has an unstretched length of 2 ft and a stiffness of $k = 50$ lb/ft.

$$F_s = k(\Delta l) = k \cdot s$$

$$\sum M_A = 0 \text{ for equilibrium}$$



$$X_1 = 3 \cos \theta \quad S = 6 \sin \theta$$

$$X_2 = 6 \cos \theta$$

$$\sum M_A = -100 - X_1 W + X_2 F_s = 0$$

$$-100 - 3 \cos \theta (200) + 6 \cos \theta (k \cdot s) = 0$$

$$-100 - 600 \cos \theta + 6 \cos \theta (50 \cdot 6 \sin \theta) = 0$$

divide by 50

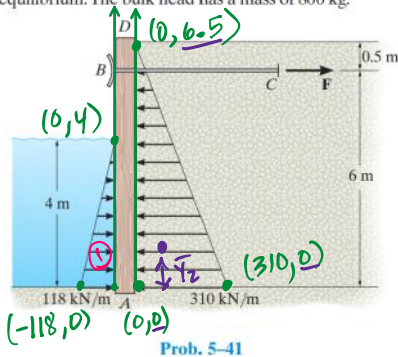
$$-2 - 12 \cos \theta + 36 \cos \theta \sin \theta = 0$$

$$-1 - 6 \cos \theta + 18 \cos \theta \sin \theta = 0$$

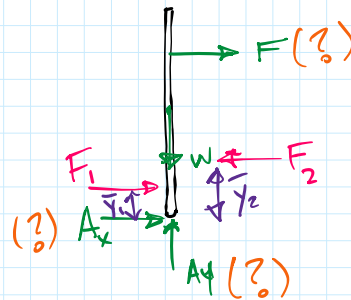
Solve graphically

$$\theta = 0.404 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right) = 23.2^\circ$$

5-41. The bulk head AD is subjected to both water and soil-backfill pressures. Assuming AD is "pinned" to the ground at A, determine the horizontal and vertical reactions there and also the required tension in the ground anchor BC necessary for equilibrium. The bulk head has a mass of 800 kg.



Ch.4 Simplify the loading



$$\sum F_y = 0$$

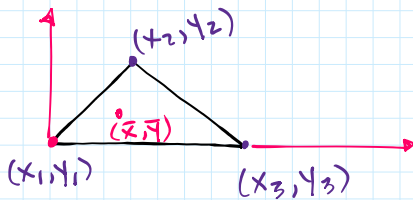
$$A_y = W = mg = 800(9.81) = 7848 \text{ N}$$

$$\sum M_A = 0$$

let you solve for F

$$\sum F_x = 0$$

let you solve for Ax



$$\bar{x} = \frac{x_1 + x_2 + x_3}{3}$$

$$\bar{y} = \frac{y_1 + y_2 + y_3}{3}$$

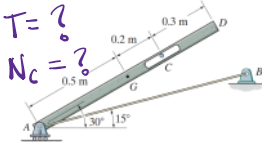
$$\bar{y}_2 = \frac{0 + 0 + 6.5}{3} = 2.16 \text{ m}$$

$$\bar{y}_1 = \frac{0 + 0 + 4}{3} = 1.33 \text{ m}$$

$$F_1 = \frac{1}{2} (4 \text{ m}) (118 \text{ kN/m}) = 236 \text{ kN}$$

$$F_2 = \frac{1}{2} (6.5 \text{ m}) (310 \text{ kN/m}) = 1007.5 \text{ kN}$$

F5-5. The 25-kg bar has a center of mass at G. If it is supported by a smooth peg at C, a roller at A, and cord AB, determine the reactions at these supports.



Prob. F5-5